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Port Digital Twins for Sustainable Urban Futures in Europe

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Abstract

Ports are increasingly recognized as actors that influence the sustainability of urban environments due to their spatial footprint, operational intensity, and close interaction with surrounding cities. As digital technologies become more embedded in infrastructure management, Digital Twins (DTs) are emerging in port systems as tools that can support more integrated and sustainable port–city development. This paper investigates how DT technologies applied in ports can contribute to broader urban sustainability objectives within port–city systems. The analysis is based on a synthesis of documented DT practices from selected European ports. Geographic Information System (GIS) visualization is used to illustrate the spatial relationship between port infrastructure and the surrounding urban environment, as well as to map the connections between DT application fields and relevant Sustainable Development Goals (SDGs). A comparative interpretation of the extent to which DT applications align with urban sustainability goals across the examined ports is achieved through the development of an SDG contribution scale. Insights derived from the European cases are subsequently contextualized for the Port of Piraeus, exploring how similar DT approaches could support both operational efficiency and the long-term climate resilience of the port–city environment. Overall, the findings provide practical insights for port authorities, urban planners, and policymakers seeking to align digital transformation strategies with sustainable and climate-responsive infrastructure development in port–city systems.

Keywords: port Digital Twins; urban sustainability; Sustainable Development Goals (SDGs); climate resilience; Geographic Information Systems (GIS); European ports; Port of Piraeus

1. Introduction

Ports are mainly located within or near urban environments and often build a port–city interface characterized by intense spatial interactions and land reclamation conflicts [1,2]. They serve as critical multimodal nodes, connecting maritime and terrestrial transport systems and forming the backbone of global trade logistics [3]. Their development involves a dual process of spatial and functional evolution, reflecting the dynamic nature of both port operations and urban growth [4]. As highlighted by the study of [5], multiple dimensions of port–city interaction can be examined. Indicative examples include the ongoing struggle over limited space [1], the spatial expansion of urban built-up areas toward port zones [6], transformations in the urban network structure [7], and broader impacts on urban economic development [8].

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Given their significant spatial, economic, and environmental footprint, ports play a pivotal role in shaping the sustainability trajectory of urban systems [9] and supporting broader global initiatives such as the Sustainable Development Goals (SDGs) [10], thus contributing to the construction and maintenance of sustainable cities and communities, and urban resilience [11]. Recent research highlights the need for ports to integrate sustainability principles into their development strategies [9]. As port–city interactions become increasingly complex, sustainability concerns among key stakeholders, including port authorities, policy-makers, users, and local communities, have intensified [9]. While ports are seen as agents of sustainable development, they also pose risks to urban well-being, particularly through the intensification of industrial emissions, land-use conflicts, and environmental degradation [1,12]. These concerns highlight the need for technological and planning approaches capable of aligning port development with long-term urban sustainability objectives.

Urban Digital Twins (UDTs) have emerged as one of the most promising innovations in urban planning and infrastructure management [13,14], with their implementation offering a valuable foundation for extending Digital Twin (DT) applications into port environments, where the urban dynamics intersect with port systems. As a specialized application of DT technology, UDTs are virtual representations of physical urban systems that integrate real-time data, enabling simulation, interaction, and synchronization [15,16]. The concept of DTs was initially introduced in the early 2000s by Grieves within the context of Product Lifecycle Management (PLM) [15,17]. Since then, the scope of DTs has evolved significantly, extending into diverse domains of urban development and leading to the development of UDTs. Today, UDTs are increasingly deployed in smart cities worldwide, serving as comprehensive digital models of physical systems, particularly in areas related to environmental sustainability [13]. Their potential is amplified through synergies with technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and Big Data (BD), which further enhance their capacity to support sustainable urban transformation [14].

As the implementation of UDTs becomes more prevalent in smart cities [18], a natural extension emerges in port cities, where port DTs should not be viewed as isolated technological applications, but rather as integral components of UDT conceptualization and implementation. In this context, Klar et al. [15] emphasize that the development of DTs for ports builds on experiences from smart cities and supply chain systems, reflecting shared technological foundations and functional interdependencies. When applied within smart ports, DT technologies offer substantial potential for enhancing logistics, navigation, automation, safety, and structural condition assessment [19–23]. While these capabilities demonstrate the growing importance of DTs in port management, their specific role in advancing the sustainability of ports and, more broadly, the sustainability of port–city systems remains underexplored [3].

Based on the above, the present paper aims to explore how port DT technologies can contribute to the sustainability of port cities and the broader port–urban interface. Emerging practices from European smart ports are synthesized to examine the evolution of ports from operational hubs into digitally integrated actors within sustainable urban systems. Port DT practices are interpreted to identify and classify the main application fields, which are subsequently linked to urban sustainability objectives aiming to understand how digital infrastructures can support the transition toward more resilient and climate-responsive port–city systems. The European examples are visualized within Geographic Information System (GIS) environments, offering spatial insight into the relationship between port infrastructure and the surrounding urban context. In addition, the identified DT application fields are mapped against relevant SDGs, enabling a comparative interpretation of how digital practices contribute to urban sustainability objectives across different port

environments. The insights gained from European cases are contextualized and applied to the Port of Piraeus, Greece, to explore potential pathways for DT integration in a Mediterranean port–city setting.

The novel contributions of this study include the following:

- Ports are reframed as digital actors within urban sustainability systems, rather than being examined solely as operational logistics infrastructures. Port DTs are explicitly linked to urban SDGs, offering an interdisciplinary perspective on port–city interaction.
- A comparative synthesis of European port DT initiatives is developed, highlighting recurring application fields and technological practices that support sustainability objectives.
- Relationship illustrations between DT application fields and urban SDGs are formed, while a preliminary Digital Twin Sustainability Index (DTSI) is introduced to comparatively represent the extent to which DT applications contribute to relevant SDGs across different ports. The DTSI is represented by an SDG contribution scale integrated into the GIS-based maps.
- Insights derived from the European cases are applied to the Port of Piraeus, providing an exploratory framework for how DT approaches could support operational efficiency and the long-term climate resilience of the specific port–urban area.

Ultimately, the practical implications of this research lie in providing actionable insights for port authorities, urban planners, and policymakers seeking to leverage digital innovations for climate-conscious and sustainable port–city futures.

2. Materials and Methods

2.1. Port Digital Twins: Conceptual Foundations and Sustainability Applications

DTs represent an innovative integration between physical systems and virtual models, continuously synchronized through real-time data and algorithmic feedback [15,16,24]. In port environments, DT systems integrate physical infrastructure, digital models, and analytical tools to support monitoring, simulation, and operational decision-making. The main components of DT systems and their relevance to port applications are summarized in Table 1.

In practice, DTs build upon existing digital tools such as Building Information Modeling (BIM) and GIS, whose functionality can be enhanced through real-time sensor data and AI systems operating within IoT architectures. These components work together to simulate and manage processes across multiple operational domains [25]. As described by [16], the core functionality of Digital Twins revolves around three key processes: mirroring, shadowing, and threading. Mirroring refers to the ability of DTs to generate a virtual representation of the physical asset, capturing both its structural characteristics and behavioral patterns. This transformation from physical to digital enables knowledge transfer across systems. Computer-aided design (CAD) tools are frequently used to support this representational function. Shadowing involves the timely monitoring and synchronization of the digital model based on changes detected in the physical system, ensuring the virtual twin remains up to date. Threading refers to the ability of DTs to connect different operational stages and subsystems, allowing for a fully integrated and interactive digital environment. This capability ensures that data and information are unified across functions, enabling a cohesive and holistic representation of the entire system.

Recent studies highlight the growing significance of DTs in port operations, with increasing investment from industry professionals and port authorities [26]. As integrated platforms for simulation, real-time monitoring, and decision-making, DTs contribute significantly to operational efficiency. Their applications in traffic prediction, infrastructure

maintenance, safety enhancement, and energy optimization may provide tangible value for port management [15], as well as for the surrounding urban systems they influence. As synthesized in [17], several studies link DT deployment to sustainability objectives, underlining their cross-sectoral relevance.

Table 1. Core components of Digital Twin systems based on [15,16,24] and their relevance to port infrastructure.

DT Component	Description	Relevance to Port Systems
Physical system	Real asset/infrastructure/operation mirrored by the digital model	Quay walls, cranes, vessels, cargo handling, etc.
Virtual model	Digital representation reflecting behavior and structure, updated through data inputs	Computational simulations enabling system design, performance assessment, and maintenance forecasting
Infrastructure and equipment	Information exchange across the digital-physical interface through sensors, IoT platforms, and monitoring systems	Unmanned Aerial Vehicles (UAVs), Non-Destructive Testing, Global Positioning System (GPS), GIS, etc.
Analytics and modelling	System monitoring, performance optimization, and predictive intervention	AI and predictive algorithms for operational and structural health monitoring, traffic and logistics optimization, and maintenance
Decision-support layer	Integration of real-time insights into management	Operational planning, coordination, and management decisions
Lifecycle Integration	Reflection of the full lifecycle of the physical system that initiates from planning and construction to operation, maintenance, and decommissioning [22]	Long-term asset management to establish climate-resilient and sustainable ports

A notable development is the connection between port DTs and the concept of Digital Twin Cities (DTCs), where urban-scale digital representations support strategic planning, improved emergency response, and enhanced energy management [27]. In port environments, DTs facilitate continuous monitoring and optimization of operations such as fuel consumption, energy use, and emissions reduction, contributing directly to sustainability objectives and regulatory compliance [28]. When integrated with BIM and GIS, DT systems also enable predictive maintenance and resource allocation, reducing unnecessary interventions, extending asset lifespans, and lowering operational costs. These functions align with the priorities of the UN's Agenda 2030 [29]. Indicatively, in the case of port infrastructure, integration with Structural Health Monitoring (SHM) methods allows DTs to anticipate degradation and prevent redundant maintenance, thus advancing long-term sustainability [24].

In addition to their efficiency-driven capabilities, port DTs are also increasingly seen as tools for decarbonization strategies, particularly given the substantial contribution of shipping to global greenhouse gas emissions. According to the International Maritime Organization (IMO), maritime transport is responsible for approximately 2.89% of global CO_2 output. In this context, the study of [27] proposed a DT framework that coordinates maritime stakeholders to assess and reduce carbon emissions. Similarly, a real-time life cycle assessment (LCA) model coupled with DT technology has been introduced by [30], simulating various fuel types and providing ongoing insights into system performance and emissions efficiency throughout a vessel's lifecycle.

As critical hubs in global trade, ports handle approximately 90% of the world's cargo and contribute significantly to urban energy consumption and reduced carbon footprint [15,31]. Their energy-intensive nature places increasing pressure to meet environmental standards while maintaining competitiveness. Accordingly, energy efficiency, air quality,

and climate resilience have become top environmental priorities, as emphasized by the European Sea Ports Organisation [15]. In this context, DTs can support the optimization of port processes, such as ship handling, terminal operations, and container transfers through simulation, machine learning, and process automation. These improvements reduce turnaround times and congestion, while supporting more sustainable and energy-efficient logistics planning.

Taken together, the sustainability-oriented gains discussed above reveal the targeted applications embedded in port functions. DT applications span cargo transportation and container storage optimization, and real-time data sharing [19]. In addition to operational logistics, DTs are applied for ship navigation safety purposes by improving docking precision and minimizing collision risks through real-time data integration [20]. They also enhance security resilience through embedded cybersecurity mechanisms, protecting from external threats [21]. Moreover, in automated terminals, DTs coordinate complex workflows by integrating inputs from GIS, BIM, and IoT sources, enabling multi-dimensional monitoring and control [22]. These diverse applications of DTs contribute to both the functional core of port operations and the broader goals of data-driven urban–port systems.

While Digital Twins offer a range of sustainability and efficiency benefits, several barriers have been identified in their implementation, linked mainly to the fact that the adoption of DTs is not solely a technical process but also a cultural acceptance [32]. One critical limitation is the current shortage of specialized personnel with the necessary expertise to deploy and operate such emerging and rapidly evolving technologies. Data acquisition also presents practical difficulties, such as ensuring sufficient storage capacity and transforming non-technical data into machine-readable formats. Within the port sector, these challenges are further amplified due to the port environment's operational complexity. As noted by [15,17], the breadth of DT applications complicates the development of a consistent framework, particularly as current approaches tend to focus on decentralized policies rather than integrated solutions. Port operations depend on the coordination of numerous actors and workflows, making system-wide DT deployment particularly demanding.

According to [15], the deployment of port DTs involves multiple technical requirements, including sensor networks, reliable data infrastructure, appropriate equipment, and trained personnel, all of which lead to high initial investment costs. This financial barrier may further widen the gap between technologically advanced ports and those with limited resources. Moreover, to ensure effective functionality, DTs require access to accurate, reliable, and continuously updated data to accurately represent real-world conditions. This is particularly critical in ports, where interdependent processes such as cargo handling, vessel movement, and inland transport generate high data volumes. In addition, DTs should be resilient to cyberattacks and comply with strict cybersecurity protocols. Beyond these technical issues, data sharing between stakeholders, though essential in multi-actor environments like ports, remains limited due to concerns over sovereignty, commercial competition, and intellectual property rights. Standardization is also a major concern. Given the dynamic nature of port systems, there is a need to harmonize technologies across data collection, analysis, and decision-making layers. Finally, port DTs must also remain flexible and adaptive to continuously evolving operational requirements.

Although the described existing framework emphasizes the efficiency-driven and operational capabilities of port DTs, it is important to acknowledge that ports themselves perform a broader multifunctional role within the urban environment. Therefore, one complementary perspective that should be considered is that ports operate as structured systems of connectivity, often described as "knots", where heterogeneous logistical, infrastructural, and governance elements are integrated to support global trade flows [33].

Moreover, ports function as urban and socio-economic interfaces, particularly through cruise and passenger activities, which introduce a more experiential and symbolic dimension, frequently associated with urban visibility, attractiveness, and tourism dynamics [34]. This dual interpretation highlights that, while current DT applications are typically developed to address specific operational domains, most commonly logistics and cargo-related functions, they often remain fragmented and purpose-specific. In this context, the multifunctional nature of ports suggests the potential value of more integrated or composite DT frameworks, capable of capturing both efficiency-driven logistical processes and the experiential and socio-economic dynamics associated with passenger and cruise activities. Such an approach could contribute to a more comprehensive representation of port–city systems and their interaction with the urban environment.

The insights discussed thus far support a comprehensive conceptualization of what constitutes a DT in the port context. As illustrated in Figure 1, an effective port DT sustainable system depends on the interaction of multiple components, including its expected benefits, application fields, operational challenges, resource requirements, and the various stakeholders involved. This multidimensional view captures the complexity of developing port DTs and highlights the need for strategic alignment with broader sustainability ambitions. In this light, relevant Sustainable Development Goals (SDGs) such as SDG 7 (clean energy), SDG 9 (industry and innovation), SDG 11 (sustainable cities), SDG 12 (responsible consumption), SDG 13 (climate action), SDG 14 (life below water), and SDG 17 (partnerships) should be actively considered during the conceptualization of port DT frameworks. Embedding these goals from the outset ensures that DT systems optimize port operations while also contributing to a healthier port–city interface and supporting long-term urban sustainability transitions.

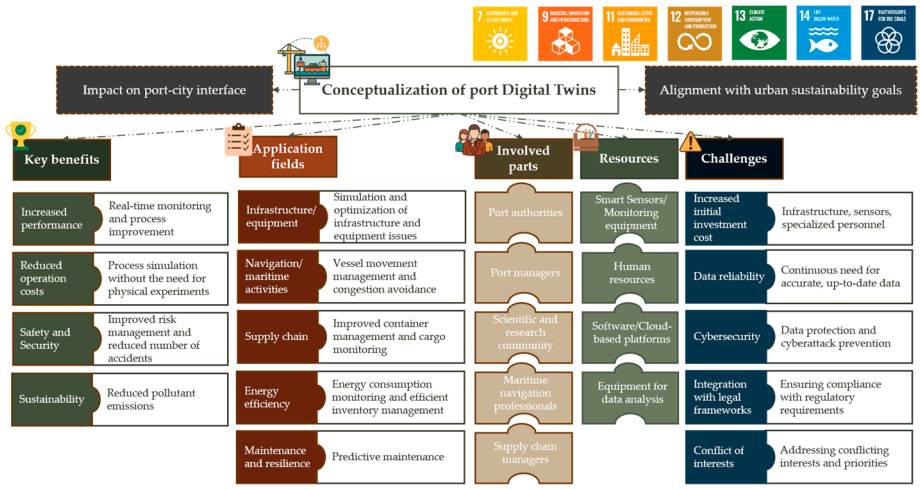


Figure 1. Conceptual illustration of the core dimensions of port Digital Twin (DT) systems, including expected benefits, application fields, involved stakeholders, required resources, and implementation challenges. The figure presents a schematic synthesis in which these dimensions are further analyzed and interpreted to illustrate their interactions and influence on DT development and application. This integrated perspective supports their alignment with broader urban sustainability objectives, particularly in relation to seven (7) Sustainable Development Goals (SDGs).

Bridging this conceptual framework with real-world implementation requires a structured and phased approach, as depicted in Figure 2. The process begins with a thorough assessment of existing practices in ports that have already adopted DTs, focusing on areas of application (e.g., navigation, infrastructure, logistics), employed technologies, and measurable outcomes. This is followed by the preparation phase, which involves the

allocation of resources, personnel training, and acquisition or development of the necessary infrastructure. Once in place, the operational deployment includes sensor installation, data collection, and system integration, followed by simulation, forecasting, and real-time decision-making. Ensuring long-term effectiveness depends on continuous monitoring, iterative improvement, and the adaptability of the DT to evolving port conditions and urban priorities.

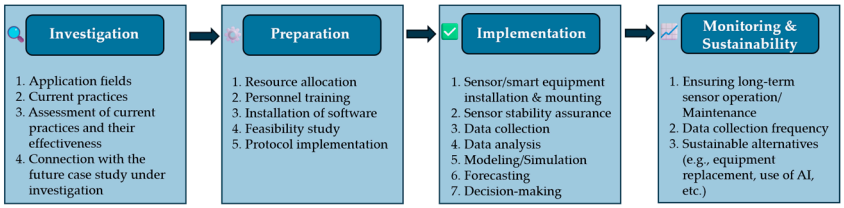


Figure 2. Stepwise general methodological workflow for the development and implementation of port Digital Twin (DT) systems synthesized based on literature review and current practices, structured into four main phases: investigation, preparation, implementation, and monitoring. The figure represents a structured framework that organizes the key stages, actions, and decision-support processes required for the deployment and continuous operation of DTs in port environments.

2.2. Methodological Framework

Following the conceptual understanding of DTs in port environments developed in Section 2.1, this section outlines the methodological steps followed to investigate how DT technologies are being implemented across European ports (Figure 3). The objective is to identify recurring application fields, technological components, and transferable practices that contribute to port sustainability and digital transformation, while also aligning with broader urban sustainability goals. The methodology was designed to allow for cross-case comparisons while remaining adaptable to the specific characteristics of different port environments. A key consideration in applying this approach is ensuring that selected case studies share comparable features, so that lessons learned and best practices can be meaningfully transferred and effectively adapted.

The methodology combines literature review, document analysis, and GIS-based visualization to support a comparative interpretation of DT implementation across different port environments.

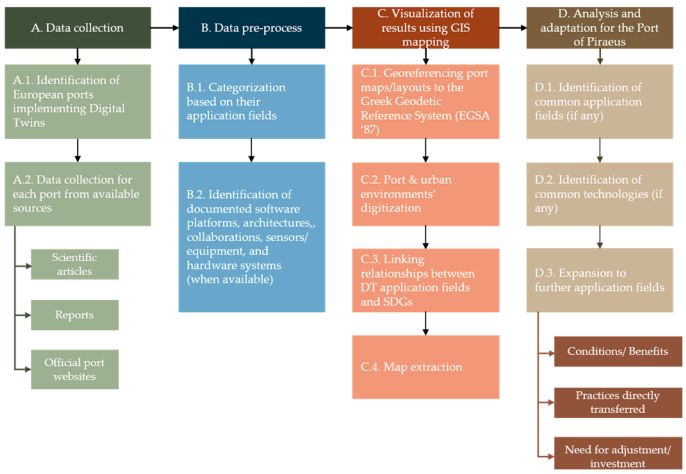


Figure 3. Methodological framework applied in this study for the identification, analysis, and transferability assessment of Digital Twin (DT) practices across port–city systems. This stepwise process followed in the present research illustrates the path from data collection and classification of DT application fields to GIS-based visualization, SDG linkage, and the development of a Digital Twin

Sustainability Index (DTSI). Insights derived from existing port practices can be systematically interpreted and adapted to other port contexts, supporting the evaluation and planning of DT implementation at the local scale, herein the case of the Port of Piraeus.

2.2.1. Phase A: Data Collection

The investigation began with the identification of European ports currently implementing DT-related practices. Data were collected from multiple sources, including scientific publications, technical reports, and official port authority documentation. For each port, available information on DT applications was compiled with attention to the systems' scope and the technologies employed.

2.2.2. Phase B: Data Pre-Processing

Following data collection, all identified DT systems were categorized based on their primary application field. For the purpose of the present research, six main fields were defined: navigation (port traffic/vessel management), container logistics and management (cargo handling and management), energy systems, infrastructure monitoring (condition and maintenance), environmental monitoring/performance, and control operation centers (cybersecurity). This categorization enabled a structured comparison of DT initiatives across the examined ports. In parallel, the resources used in each case, such as sensors, platforms, and software, were documented to better understand the technological landscape supporting DT implementations.

2.2.3. Phase C: GIS-Based Visualization

To facilitate spatial interpretation of DT applications, the identified DT fields were visualized using a GIS environment. Initially, port layouts were georeferenced using the Greek Geodetic Reference System (GGRS '87), followed by the digitization of key terminal areas. In addition, the surrounding urban environment was digitized using polygons to capture the spatial extent of the port–city interface. This allowed for an initial comparative insight into the port–city interface and the broader urban extent.

To enhance the clarity of thematic representation, each DT application field was associated with a unique vector-style icon (e.g., the earth for environmental applications or a container for logistics), allowing for an immediate visual interpretation of the dominant DT fields within each port environment. Furthermore, the identified DT application fields were linked to the relevant urban SDGs, enabling the visualization of how different DT practices contribute to broader sustainability objectives within port–city systems. The linkage between DT application fields and SDGs was established through a structured interpretative process, based on literature evidence, case applications and impacts, as well as the functional role of each DT field (e.g., logistics optimization, energy monitoring, navigation safety). Therefore, each DT application field was associated with one or more SDGs according to its primary operational impact, ensuring consistency across all case studies.

The extent to which DT applications align with these SDGs was comparatively illustrated through two complementary scales integrated into the GIS maps. The first scale reflects the number of DT application fields identified in each port, indicating the breadth of DT implementation. The second scale represents the number of SDGs associated with these DT applications, illustrating their contribution to broader sustainability objectives within port–city systems. The assessment is qualitative-comparative and all SDGs are considered with equal importance. Based on this approach, a preliminary DTSI is introduced and expressed as follows:

$$\text{DTSI} = f(\text{DT application fields, SDGs}) \quad (1)$$

The developed DTSI is considered as a heuristic and comparative tool that captures the extent to which DT applications support sustainability-related objectives. Its purpose is to enable cross-case comparison and to support the identification of potential areas for DT development within port–city systems.

2.2.4. Phase D: Analysis and Adaptation for the Port of Piraeus

The final phase focused on drawing conclusions applicable to the Port of Piraeus. At this phase, the common features across the studied European ports were identified, including recurring application fields and frequently used technologies. Based on this, a forward-looking proposal was developed for Piraeus, identifying transferable practices, possible conditions and benefits, and areas where further investment or adaptation may be required to support broader sustainability and digital transition objectives.

2.3. Cross-Case Knowledge Synthesis: European Innovative Ports

Within the scope of this research, six European ports, namely Antwerp-Bruges, Barcelona, Hamburg, Livorno, Rotterdam, and Valencia, were selected for analysis based on a combination of criteria. These include the availability of documented and consistent data on the development and implementation of DT systems, as well as the demonstrated engagement of these ports in digital transformation initiatives, research and innovation activities, stakeholder collaborations, and advanced operational practices (Figure 4). In this context, "documented" DT initiatives refer to applications that are supported by verifiable sources, such as peer-reviewed scientific literature, official port authority reports and platforms, or well-established project documentation, rather than conceptual proposals or isolated pilot studies. Therefore, the selected ports offer a balanced representation of different levels of digital maturity within the European port system and maintain comparability under a shared institutional and policy context. Considering this, while other European ports, such as Anzio and Belfast [3], Gothenburg, and Algeciras and Genoa [29], have also launched DT-related initiatives, these were not included due to limited access to sufficiently comprehensive, consistent, and verifiable data required for a robust cross-case analysis. Moreover, the selection was intentionally focused on European Union ports in order to ensure consistency in regulatory frameworks, policy environments, and sustainability priorities, which are particularly relevant when assessing DT applications in relation to SDGs and climate-oriented strategies. In the following, the six ports are alphabetically presented as innovation hubs, with a detailed review and analysis of their DT applications provided in Section 3.

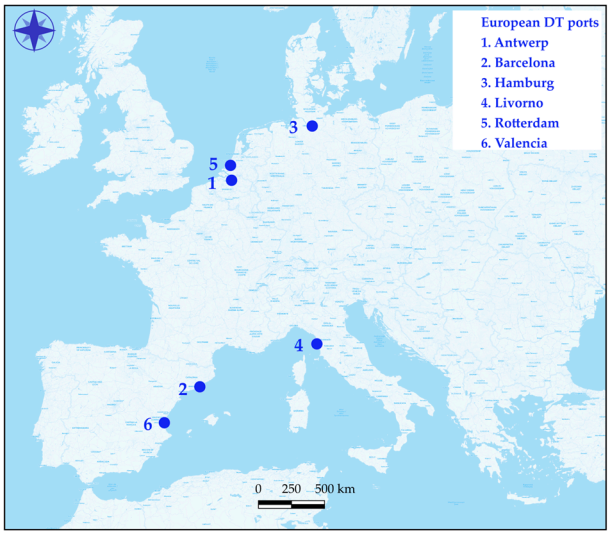


Figure 4. The six European ports with documented DT innovations presented in alphabetical order.

2.3.1. The Port of Antwerp-Bruges

The Port of Antwerp-Bruges, located within the Belgian city, is one of the key pillars of the European port system, ranking as the second largest in Europe after the Port of Rotterdam. Today, the port maintains connections with over 800 destinations and has a considerable economic significance is also considerable, contributing more than €20 billion in added value to the region [35].

The port's operations span the handling and storage of a wide range of cargo types. Its primary functions include ship loading and unloading, warehousing, and further distribution of goods [36]. Its facilities comprise over 6 million square meters of storage space. The port occupies 14,000 hectares, including 1992 hectares of water bodies, and it operates under a landlord model, wherein land is leased to private terminal operators. From the late 1950s to the early 1960s, several shipping companies established proprietary terminals in the area, marking a shift in the port–city interface. The growing size of vessels, the rise in containerization, and technological advances drove terminal operators to seek berths with greater draft capacity [37].

The Port of Antwerp-Bruges is steadily evolving as a hub for smart and digital port infrastructure, implementing a wide array of innovations aimed at improving operational efficiency, safety, and sustainability. Investments have been made in advanced systems supporting traffic management, including the collaborative logistics platform Bulkchain in partnership with NxtPort, as well as the deployment of digital cameras and smart sensors, such as iNoses to measure air quality [38]. Moreover, in collaboration with the technology company Zensor, the port has enhanced mooring safety by installing digital sensors on bollards to measure tension forces [39]. In addition, the port participates in the PIONEERS project, which focuses on managing and reducing emissions in port operations [40]. These implementations collectively reflect Antwerp's broader strategy of becoming a digitally integrated hub.

2.3.2. The Port of Barcelona

The Port of Barcelona stands as one of the most diversified ports in the Mediterranean, hosting over 35 specialized terminals for container handling, vehicle logistics, and both solid and liquid bulk. It is also home to the Logistics Activities Zone (ZAL), recognized as one of southern Europe's most efficient logistics platforms. Its maritime services extend to more than 200 global destinations, while inland rail links offer regular services to major Spanish cities and key European locations [41].

Since the early 1980s, strategic efforts have been aimed at enhancing the interface between the port and the city. A pivotal urban planning initiative was commissioned in 1981 to restructure the quay and surrounding urban landscape, later leading to the establishment of the Port Vell Urban Development Authority in 1988. The vision focused on retaining the port's character while diversifying land use and improving integration with the urban environment. Today, the Gerencia Urbanística Port Vell continues to oversee redevelopment initiatives, promoting adaptive reuse and sustainable urban–port interactions. As part of its southern expansion, the port developed a second access point—Bocana Nord—creating new waterfront areas. To manage the planning and land use of this area, the Special Plan for Nova Bocana was adopted in 2001, with successive updates in 2006, 2011, and 2018. These urban regeneration efforts reflect the port's evolving role as both a logistical engine and a dynamic part of the city's development strategy [41].

In recent years, the Port of Barcelona has emerged as a leading example of digital transformation in the Mediterranean, driven by a strategic commitment to sustainability, efficiency, and technological innovation. Guided by the SMART (Sustainable, Multimodal, Agile, Resilient, and Transparent) framework, the port has launched multiple initiatives to enable real-time monitoring, automate logistics operations, and enhance supply

chain visibility [42]. Central to this digital transition is the integration of DT technology, particularly in container and logistics management [43].

Barcelona's innovation strategy is further supported through collaborations with tech start-ups. For example, 3D Modelling Studio leverages DT applications to classify waste and improve environmental oversight [42], while AllRead Machine Learning Technologies has developed an AI-based computer vision system that enables real-time container identification by already analyzing over 950 trains and 13,500 containers [44]. Optical Character Recognition (OCR) tools are also incorporated to enhance the speed and accuracy of container handling from trucks. Additional innovations include the pilot adoption of LoRaWAN IoT networks tested alongside GPS [42]. The OpenPort initiative exemplifies the port's open innovation model by fostering synergies between start-ups and the port-logistics community, while the Ports 4.0 fund supports the creation and scaling of startups, spin-offs, and new tech-driven business models oriented toward market-ready solutions [41].

Operational innovations further include the use of aquatic drones for remote inspections, zero-emission vans, autonomous buses, and collaborative projects with startups like goClever, which is testing smart sensor systems to optimize waste management. Meanwhile, the 5G Maritime project enables precise, real-time geolocation to enhance vessel coordination [41]. Collectively, these initiatives reflect Barcelona's commitment to integrating innovation with sustainability objectives.

2.3.3. The Port of Hamburg

The Port of Hamburg is Germany's largest and most versatile seaport, handling over 126 million tons of cargo annually, including nearly 8.5 million TEUs. Its strategic location between the North and Baltic Seas, combined with extensive multimodal transport links, enables seamless connections to Central, Eastern, and Northern Europe. With more than 75 terminals, including state-of-the-art container facilities and multipurpose terminals for heavy and breakbulk cargo, the port plays a vital role in national and European logistics. It generates €20 billion in annual gross value added and supports over 260,000 jobs, affirming its role as a cornerstone of economic activity [45].

In parallel with its port functions, Hamburg has pursued an integrated urban development strategy. HafenCity, one of Europe's most ambitious waterfront regeneration projects, is transforming 160 hectares of former port land into a vibrant urban district. The port's evolution is guided by the Port Development Plan 2040, which sets a long-term vision for modernization and innovation while reinforcing Hamburg's status as a maritime and logistics hub. The broader Hamburg Metropolitan Region further strengthens the port's value, encompassing four federal states and a collaborative governance framework. With major industries such as aviation, shipbuilding, and renewable energy, alongside digital logistics and AI innovation, the region fosters synergies between infrastructure, research, and economic growth. Cross-border initiatives, funded through regional cooperation programs, support sustainable mobility, clean energy, and port–city integration [45].

Described as a "smart knot" that integrates governance, management, and logistics into a coordinated system of connectivity [33], while simultaneously supporting tourism and cruise-related activities as part of its urban interface [34], the Port of Hamburg is at the forefront of digital transformation, leveraging DT technologies integrated with 5G infrastructure to enhance operational efficiency, environmental monitoring, and port logistics. Through the MoNArch 5G architecture, the port has implemented three core applications: (1) low-latency communication for intelligent traffic light control; (2) high-bandwidth AR support for asset management; and (3) massive machine-type communication to collect environmental data from barge-mounted sensors. These components enable

real-time data synchronization across DT systems and support proactive traffic and logistics coordination. The DT platform aggregates information from diverse sources—sensors, mobile units, and stakeholder databases—enhancing traffic planning, particularly for container truck flows, and providing high-resolution geolocation services. The system also integrates VR/AR tools to visualize port operations in real-time, supporting informed decision-making and predictive maintenance strategies [19].

Complementing these efforts, the EUROGATE terminal is developing a specialized DT for container handling, utilizing IoT data from gantry cranes and straddle carriers. This data feeds into simulation models that optimize equipment dispatch and storage allocation, balancing performance with environmental objectives [17]. Further extending its digital leadership, the Port of Hamburg co-founded the international "chainPORT" network—a collaborative platform connecting global ports such as the ports of Antwerp-Bruges, Barcelona, and Rotterdam, discussed in this section. This initiative fosters innovation and shared development of future-oriented smart port solutions [45].

2.3.4. The Port of Livorno

The Port of Livorno, located on Italy's western coast along the Tyrrhenian Sea, plays a strategic role as Tuscany's main maritime gateway. The port benefits from its central Mediterranean position and excellent multimodal connections. As a key node in the TEN-T Core Network (Scandinavian-Mediterranean Corridor), Livorno ranks among Italy's top-performing ports, handling over 35 million tonnes of cargo and more than 3.5 million passengers annually, including significant cruise traffic. Spanning over 2.5 million square meters, the port plans for further expansion to −20 m under the ambitious Darsena Europa project. This major redevelopment initiative will also relocate environmentally impactful activities away from the historic port, freeing up space and improving sustainability [46].

The Port of Livorno is currently undergoing a significant digital transformation aimed at modernizing its Information and Communication Technology (ICT) infrastructure and enhancing operational efficiency. Departing from monolithic and difficult-to-scale legacy systems, the port has introduced the MONICA Standard Platform, a modular and cloud-based ICT architecture that promotes flexibility and innovation. This platform, structured across the three canonical layers—data, business logic, and applications—enables agile service development for logistics stakeholders. It is supported by a dedicated fiber-optic network, which provides comprehensive connectivity across terminals and gates, while also addressing digital access disparities in nearby urban areas. Complementing this infrastructure, a wireless backbone delivers 100 Mbps connectivity throughout the maritime zone [47].

Since 2015, the port of Livorno has served as a real-world testing ground for next-generation solutions, in collaboration with the National Inter-University Consortium for Telecommunications (CNIT). Within the framework of the EU Horizon 2020 COREALIS project, Livorno has embraced digital transformation to enhance competitiveness, efficiency, and sustainability in port operations. A core focus has been the deployment of 5G-enabled IoT infrastructure, allowing for real-time data collection and management via smart sensors, AI, and edge computing. This supports the RTPORT module, which coordinates container handling and logistics decisions dynamically. The data feeds a digital twin engine, offering a virtual and navigable simulation of the terminal environment, enabling operators to monitor and optimize operations with unprecedented precision [31]. Livorno's COREALIS engagement reflects its ambition to become a forward-looking, intelligent, and sustainable port of the future.

2.3.5. The Port of Rotterdam

The Port of Rotterdam serves as the maritime gateway to continental Western Europe and has a long-standing history dating back to the mid-15th century. Its development is closely interwoven with the economic, industrial, and commercial evolution of Europe. Today, the port spans 40 km, accommodating approximately 30,000 seagoing vessels and 130,000 inland barges annually. In 2022, it handled over 467 million tonnes of cargo, including 14.4 million TEUs, ranking it fifth globally in bulk throughput and tenth in container handling [37].

The Port of Rotterdam is actively advancing its transformation into a smart port by integrating digital technologies and intelligent infrastructure. Central to this initiative is the deployment of smart sensors, AI, and Machine Learning (ML) to support inspection, maintenance, and lifecycle management. Notable innovations include the use of smart bollards that measure mooring line tension to ensure safe berthing, as well as a floating autonomous sailing laboratory designed to test and refine self-navigating vessel technologies—part of the port's broader ambition to accommodate autonomous ships by 2030 [48].

The port collaborates with technology leaders such as IBM, Esri, Cisco, and Axians to develop an integrated IoT platform. Hundreds of installed sensors feed real-time data into this system, while augmented reality (AR) and smart weather data enhance operational awareness [49]. The result is optimized scheduling of vessel arrivals and departures and even the facilitation of ship-to-ship cargo transfers without the need for berthing [50,51]. Moreover, leveraging IoT and AI, the port gathers meteorological and hydrological data, such as salinity, wind speed, currents, and tides, in real time [17,38]. This information contributes to the forecasting of port conditions, such as visibility and docking feasibility on a given day [19]. OCR systems are being used for container identification and optimal placement [40]. Crane and truck operations are synchronized with incoming vessel data, while communication with ships provides real-time information on location, performance, and energy use. These integrations improve hinterland connectivity and logistics node coordination [52].

Two further Research and Development (R&D) initiatives emphasize the port's innovation leadership. The RAMLAB (Rotterdam Additive Manufacturing Lab) operates within the port's shipyards, using robotic arms to 3D-print metal ship components—offering cost-effective and sustainable maintenance solutions [49,51]. Additionally, BlockLab, launched in 2017, explores blockchain applications in logistics through a joint city-port research partnership [53]. Finally, start-up ecosystems, such as PortXL and Port Innovation Lab, actively collaborate with logistics and maritime firms like Vopak and Boskalis. Their goal is to co-develop innovative solutions that support smarter and more sustainable supply chains [49].

2.3.6. The Port of Valencia

The Port of Valencia serves as a central node in the Western Mediterranean logistics network, exerting influence over a hinterland of approximately 350 km. This region accounts for over half of Spain's Gross Domestic Product (GDP) and includes nearly 50% of the national labor force. With efficient connectivity extending across southern Europe and North Africa, the port of Valencia supports trade operations that impact a market of roughly 270 million people. The port is equipped to handle high volumes of cargo and also maintains maritime connections with the Balearic Islands and Italy. In recent years, cruise activity has also seen a consistent upward trend [54].

To accelerate its digital transition, the Port of Valencia has introduced a variety of smart systems and technologies across multiple operational layers. These include solu-

tions for cargo tracking, supply chain monitoring, and digital administration [43]. A cornerstone project in this transformation is TwinShip aiming to develop an open-source digital twin framework utilizing AI and machine learning to support emissions reduction in shipping operations. With targets of 30–40% emissions reduction by 2030 and net-zero goals by 2045, TwinShip integrates clean technologies and autonomous vessels, offering a data-driven decision-support system for fuel optimization and regulatory compliance [55].

In terms of digital governance, the ValenciaportPCS platform plays a central role. It acts as a secure, neutral interface for public and private actors to exchange data and automate logistics procedures [56]. Moreover, innovation at the Port of Valencia is further supported through R&D partnerships and startup collaborations. One such initiative is OPENTOP, an open innovation hub designed to pilot and scale technological solutions for port-related challenges [57]. In parallel, the port actively engages in the Ports 4.0 program, which funds innovation from early ideation to industrial implementation. This initiative embraces a wide range of technologies, from AI, IoT, and Blockchain to Augmented Reality, Robotics, and Cybersecurity, addressing both disruptive innovations and process-based improvements [58].

Based on the above Table 2 summarizes the main characteristics of the six European ports examined in this study and provides a comparative overview of the DT-related initiatives and technological domains currently supporting their digital transformation.

Table 2. Comparative overview of the main characteristics and DT-related innovation domains across the six European ports examined in this study.

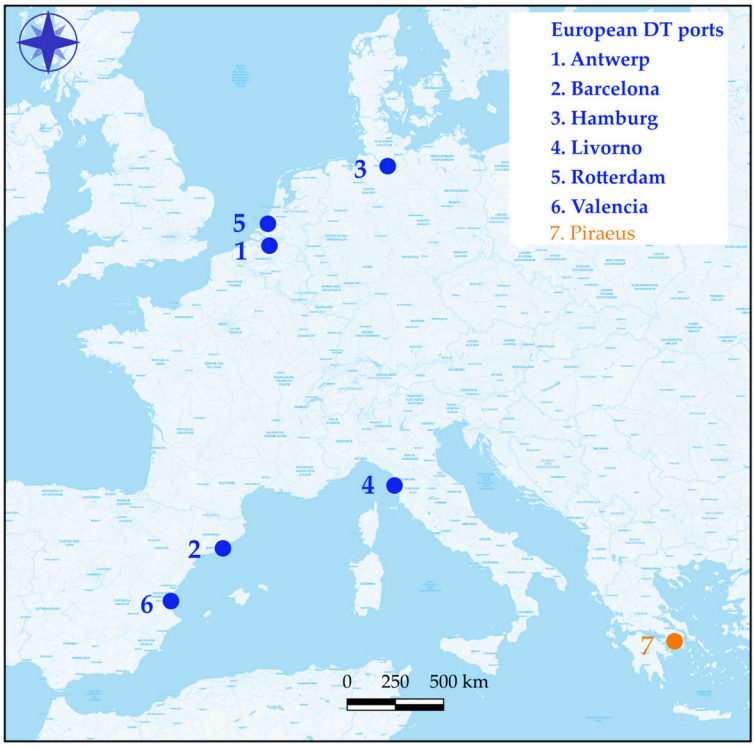
Port	Geographic Context	Key Port Characteristics	Main Innovation Domains
Antwerp-Bruges	Belgium—North Sea	Second largest port in Europe; ~14,000 ha port area; >800 global connections; major logistics hub	Logistics platforms (Bulkchain, NxtPort), air-quality sensors (iNoses), smart mooring sensors, emissions monitoring (PIONEERS)
Barcelona	Spain—Western Mediterranean	Diversified port with >35 terminals; strong multimodal connectivity; integrated urban–port development	SMART framework, AI container recognition (AllRead), OCR to improve container handling, IoT networks (LoRaWAN), smart waste monitoring, collaborative projects (goClever startup), 5G maritime connectivity
Hamburg	Germany—North Sea gateway	Germany's largest port; >126 Mt cargo/year; strong hinterland connectivity; HafenCity urban redevelopment	5G-enabled DT platform (MoNArch 5G architecture), DT for container transportation, VR/AR port operations monitoring, container terminal digital twins integrating IoT data (EUROGATE terminal), co-founded the international "chainPORT" network
Livorno	Italy—Tyrrhenian Sea	Strategic TEN-T Mediterranean corridor port; ~35 Mt cargo/year; Darsena Europa project expansion	MONICA Standard Platform—a modular and cloud-based ICT architecture for logistics, 5G IoT sensors, RTPORT logistics platform, digital twin simulation of terminal operation
Rotterdam	Netherlands—North Sea	Europe's largest port; ~467 Mt cargo/year; advanced smart port infrastructure	Collaboration with technology leaders such as IBM, Esri, Cisco, and Axians to develop an integrated IoT platform, AI-driven logistics optimization, predictive environmental monitoring, OCR systems for container management, BlockLab

			through a joint city-port research partnership, start-up ecosystems, such as PortXL and Port Innovation Lab
Valencia	Spain—Western Mediterranean	Major Mediterranean logistics hub; strong Iberian hinterland connectivity	Cargo tracking platforms, TwinShip to support emissions reduction in shipping operations, digital port community systems (ValenciaportPCS), AI logistics tools

2.4. The Port of Piraeus

The selection of European ports as reference cases for DT applications is grounded in their geographic proximity, regulatory alignment, and cultural compatibility with the Port of Piraeus. European ports operate within a shared legislative framework, such as the EU Port Services Regulation [59] , which governs the provision of port services, and the TEN-T (Trans-European Transport Network), which sets priorities and financing mechanisms for the development of strategic transport infrastructure, including core ports. These conditions facilitate the transferability of technological know-how, especially through EU-supported initiatives such as Horizon and CEF programmes, providing a pathway for applying European innovation frameworks to the Greek context.

The Port of Piraeus is the largest and most significant maritime hub in Greece (Figure 5). Located at the interface of Europe, Asia, and Africa—just a few kilometres from Athens—it serves as the main national gateway for passengers and cargo, linking the Greek mainland with its islands and supporting international cruise and trade traffic. Due to its strategic location along the Mediterranean corridor, Piraeus functions as a key node in global supply chains, facilitating intermodal connectivity with the Balkans and the Black Sea. The port is part of the TEN-T Core Network and ranks among the top European ports in terms of traffic volume. It is Europe's largest passenger port and holds the seventh position in container throughput among EU ports [60].



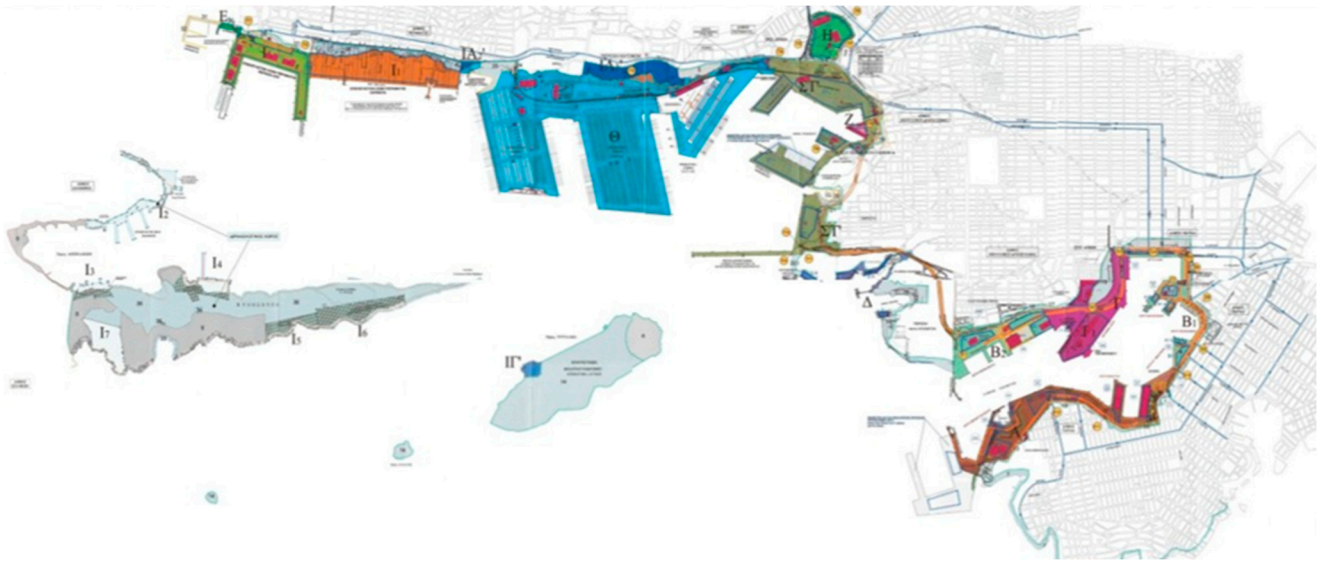


Figure 5. The Port of Piraeus: Geographical location within the European network and schematic representation of its Master Plan based on [61].

Since the acquisition of a 67% stake in Piraeus Port Authority S.A. (PPA S.A.) by China COSCO Shipping Corporation in 2016, the port has entered a phase of expansive growth. This is reflected in the port's ten-year Master Plan and Development Program, which includes significant investments across cargo, cruise, ship repair, and logistics infrastructure [60]. The port comprises specialized terminals for container handling, car and general cargo transport, free zone operations, coastal and cruise shipping, as well as shipyard services [60,62].

Environmental sustainability and digital innovation form a central part of PPA's strategic outlook. The port implements an Integrated Quality, Environmental, and Energy Management System, certified under ISO 9001:2015, ISO 14001:2015, and ISO 50001:2018 [63–65]. Continuous improvement in environmental performance, health and safety, and energy efficiency remains a strategic priority. In this context, the port has adopted various initiatives under European programs such as CEF, Horizon 2020, and INTERREG, positioning itself as a driver of sustainable port development [60].

Piraeus is also actively involved in research and innovation activities, collaborating with national and international partners to explore new technologies and operational models. Through its participation in the European EcoPorts Network, the port commits to regular self-assessment and sustainability reporting using the SDM framework. Additionally, the port implements advanced monitoring systems for marine water quality, sediment, air and noise pollution, and has developed emergency response plans for land and marine incidents involving hazardous materials [62].

Although Piraeus has yet to implement a full-scale operational DT, it is considered among the leading European ports in terms of digital governance maturity. According to the study of [43], it ranks just behind Valencia and alongside Barcelona and Algeciras, reflecting its strong administrative digitalization and ongoing modernization efforts.

3. Results and Discussion

To analyze the interconnection between port DT applications and the sustainability of the broader urban environments they serve, the methodology outlined in Figure 3 identified six European ports with active DT implementations. These ports, previously described in Section 2.3, were selected based on documented evidence from the academic literature and official port sources. In the context of that prior analysis, each port's inno-

vation ecosystem, comprising documented platforms, partner collaborations, and advanced systems, was considered part of a broader digital transformation strategy. For each case, DT applications were categorized according to their functional field (e.g., logistics, energy, navigation, infrastructure monitoring), allowing for a structured thematic analysis.

To visually represent this information, the official masterplans or terminal layouts of each port were digitized, along with the urban environment of the city in which the port operates. The urban boundary for each city was delineated using Google Maps, ensuring a uniform and comparable baseline across all port–city pairs. While these polygons reflect the administratively defined city boundaries (based on the official name), it is important to note that the actual area of urban influence may be broader in terms of socio-economic and environmental interactions. Nonetheless, this consistent approach allows for a balanced comparative analysis across diverse urban contexts.

The digitization was conducted in a GIS environment, enabling a georeferenced visualization of each port and its respective city. The identified DT application areas are not depicted with exact spatial boundaries (which are typically unavailable) but rather are represented through thematic vector-based icons placed on the port layout. This approach enhances visual clarity and facilitates cross-case comparison of DT fields. In the following subsections, each European port is individually presented through this visual framework before progressing to the conceptualization of DT applications in the Port of Piraeus.

3.1. Insights from European Port DT Applications

The Port of Antwerp-Bruges exemplifies a robust and structured integration of DT technologies across several operational domains, supporting both port performance and urban sustainability. As one of Europe's leading ports, it implements advanced DT solutions for navigation, logistics, and energy coordination, which collectively contribute to improved efficiency, lower environmental impacts, and enhanced city-port synergies. The key DT application fields are (Figure 6):

- Port traffic and vessel management: The port employs a geographical digital twin integrating Automatic Identification System (AIS), radar, weather data, and BIM to enhance real-time vessel tracking, ship berthing, and route prediction ([17,38]. The use of automated drone networks and anomalous ship movement detection further supports situational awareness and safe navigation [3]. This application field is linked to five SDGs: SDG 9 (innovation and intelligent infrastructure for vessel coordination), SDG 12 (avoiding congestion and optimizing resource use), SDG 13 (emission reduction through predictive traffic control), SDG 14 (marine ecosystem protection through controlled routing), and SDG 17 (stakeholder engagement in coordinated maritime operations).
- Logistics, cargo handling, and transportation: DT applications are used to optimize cargo flow, including dockside cargo authentication, container tracking, and supply chain balancing [38]. These systems enhance logistical efficiency and reduce idle time. This application field is linked to four SDGs: SDG 9 (technological advancement in multimodal logistics), SDG 11 (support for efficient urban freight mobility), SDG 12 (reduction in waste in handling processes), and SDG 17 (collaborative logistics platforms).
- Energy systems: The port incorporates real-time energy consumption and supply chain data into its DT environment to support load balancing and minimize environmental impact [38]. Predictive models help align port activities with local and regional energy demands. This application field is linked to three SDGs: SDG 7 (integration of clean and efficient energy solutions), SDG 11 (urban energy sustainability alignment), and SDG 12 (responsible consumption and optimization of energy use).

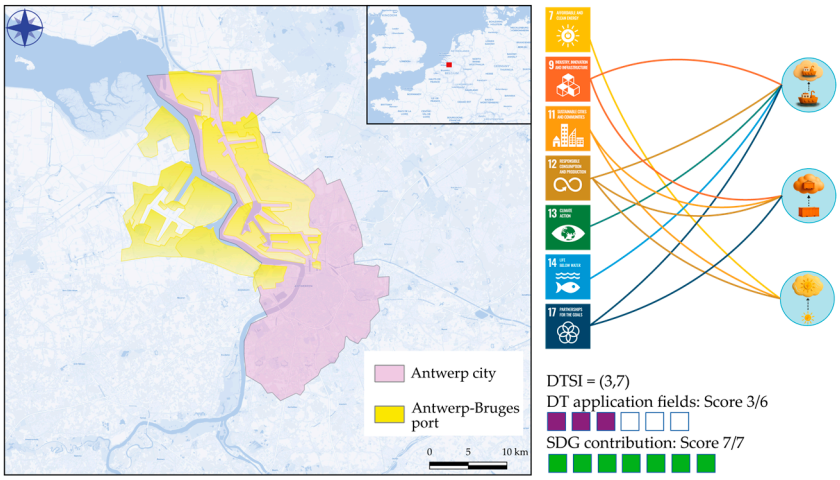


Figure 6. Digitized layout of the Port of Antwerp-Bruges and its urban interface—DT application fields and their connection with SDGs.

These application fields are directly relevant to Antwerp's urban setting, where dense population, industrial activity, and port operations converge. DT systems reduce logistical bottlenecks, lower environmental burdens, and improve spatial planning at the port–city interface, thus reinforcing Antwerp's capability to be both a technological port hub and a climate-conscious urban region. In summary, the Port of Antwerp-Bruges demonstrates activity in three of the six identified DT application fields and contributes to all seven SDGs considered in this study.

The key DT application fields of the Port of Barcelona are (Figure 7):

- Logistics and cargo handling operations and transportation [29]: Key systems include intelligent surveillance platforms for situational awareness [3] while these digital innovations contribute to four SDGs: SDG 9 (innovation in logistics), SDG 11 (sustainable urban connectivity), SDG 12 (responsible cargo handling), and SDG 17 (collaborative innovation ecosystems with start-ups and technology partners).
- Environmental monitoring: The 3D Modelling Studio startup supported by the Port of Barcelona has initiated a DT solution for measuring and classifying waste [42]. This application field is linked to SDG 11 (urban environmental quality), SDG 14 (marine protection), and SDG 17 (collaborative innovation ecosystems with start-ups).

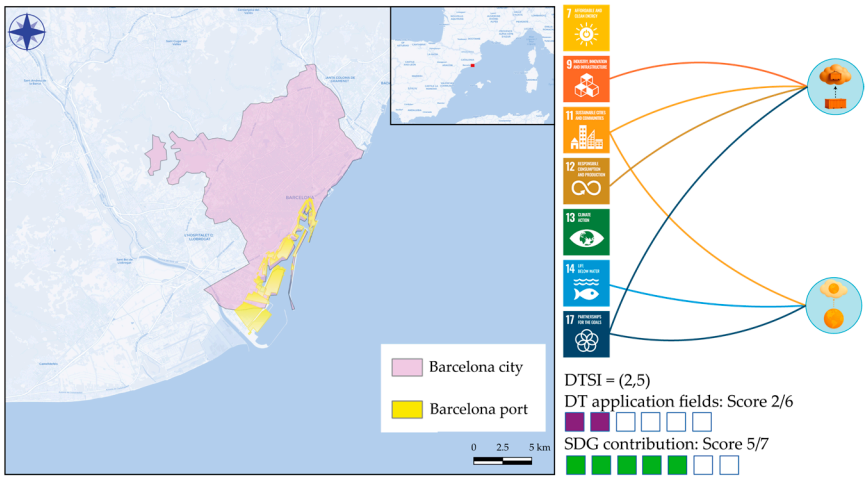


Figure 7. Digitized layout of the Port of Barcelona and its urban interface—DT application fields and their connection with SDGs.

Given Barcelona's dense urban environment and its strategic position within the Mediterranean corridor, these innovations support the city's broader sustainability agenda. The port's intelligent logistics systems streamline operations, reduce emissions, and improve coordination with inland transportation networks, reinforcing Barcelona's attempt at integrated urban–port development. In summary, the Port of Barcelona demonstrates activity in two of the six identified DT application fields and contributes to five of the seven SDGs considered in this study.

The Port of Hamburg stands as one of Europe's most advanced digital seaports, showcasing an extensive integration of DT technologies across operational, infrastructural, and environmental fields. Its role as a logistical and industrial gateway between Germany, Scandinavia, and Central Europe reinforces the strategic importance of its digital transformation in both port performance and urban sustainability. The key DT application fields are (Figure 8):

- Port traffic and vessel management: Hamburg utilizes DT platforms to manage port traffic and vessel scheduling. and predictive traffic forecasting [3,62] to minimize congestion and enhance navigational safety. This application field is linked to five SDGs: SDG 9 (real-time, intelligent infrastructure), SDG 12 (avoiding unnecessary delays and energy consumption), SDG 13 (proactive emissions control), SDG 14 (reduced marine impact through traffic planning), and SDG 17 (coordination between port stakeholders).
- Logistics, cargo handling, and transportation: DT applications enable coordination of cargo flows, with monitoring of container movement across terminals [3,29]. This application field is linked to four SDGs: SDG 9 (logistics innovation), SDG 11 (efficiency in urban freight mobility), SDG 12 (resource optimization), and SDG 17 (integrated supply chain collaboration).
- Infrastructure condition and maintenance: Hamburg's DT testbed monitors critical structures, including roads and bridges connecting the port with the city. Sensors are used to track structural fatigue, vibrations, and load impacts [66], supporting predictive maintenance strategies [19]. This application field contributes to SDG 9 (resilient infrastructure), SDG 11 (safe and sustainable connectivity), SDG 12 (efficient asset use), and SDG 13 (climate-adaptive maintenance).
- Environmental monitoring: Sensors mounted on mobile barges and across port facilities measure environmental quality [19], supporting environmental reporting and compliance. This application field is linked to SDG 11 (urban environmental quality), SDG 14 (marine protection), and SDG 17 (for potential collaborative innovation ecosystems with start-ups and other partnerships).
- Control and operations centers: The Port of Hamburg is developing a comprehensive virtual representation of its control centers within the Digital Port Twin, enabling 3D visualization and reproduction of digitalized processes for optimization purposes [67]. The deployment of physical 5G towers enhances communication within this system through beam configuration and transmission optimization [17,19]. This application field supports SDG 11 (urban–port coordination) and SDG 17 (institutional cooperation and data sharing).

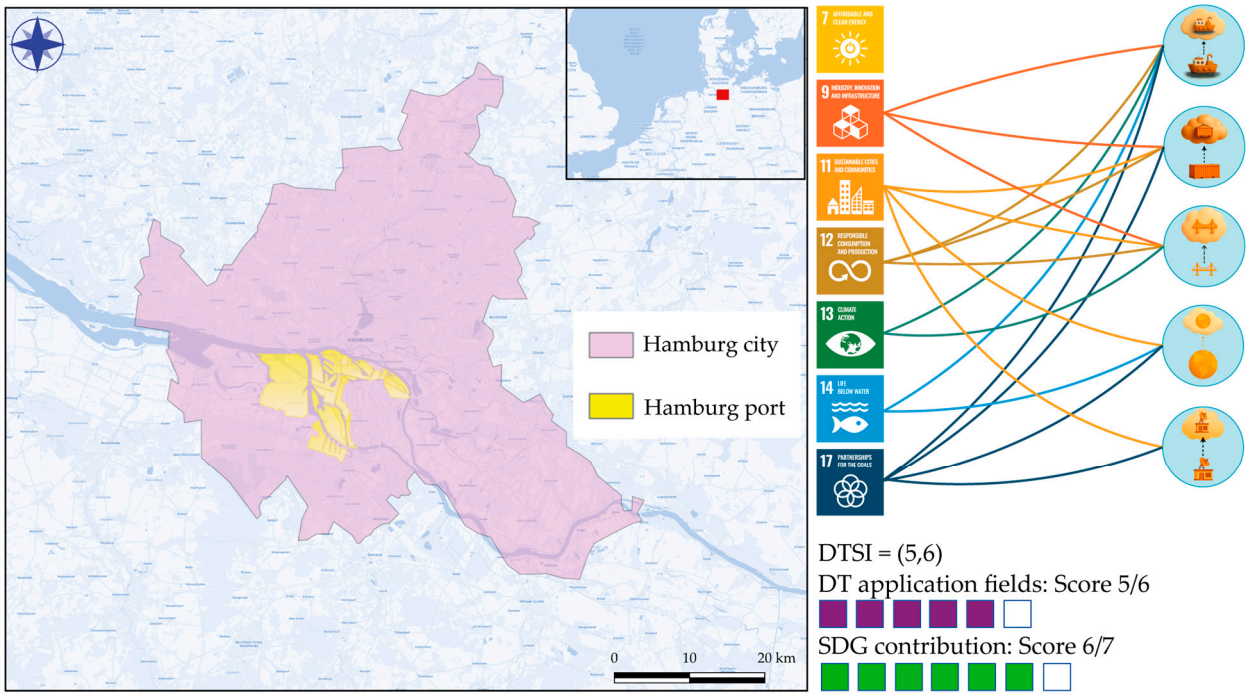


Figure 8. Digitized layout of the Port of Hamburg and its urban interface—DT application fields and their connection with SDGs.

The Port of Hamburg's comprehensive DT framework contributes to port efficiency and the livability and environmental health of the surrounding metropolitan region. With HafenCity representing one of Europe's largest urban redevelopment projects, the alignment between port digitization and urban sustainability goals becomes increasingly critical. In summary, the Port of Hamburg demonstrates activity in five of the six identified DT application fields and contributes to six of the seven SDGs considered in this study.

The Port of Livorno has made significant strides in applying DT technologies to modernize port logistics and energy coordination. The key DT application fields are (Figure 9):

- Logistics, cargo handling, and transportation: DT applications in Livorno support real-time monitoring of yard vehicles, such as cranes and forklifts, using GPS and low-latency 5G communication [47]. These systems enhance cargo positioning, handling, and flow coordination within container terminals [31]. This application field contributes to SDG 9 (digitalization of cargo operations), SDG 11 (urban mobility and logistics integration by improving freight coordination), SDG 12 (resource efficiency via reduced idle time and streamlined workflows), and SDG 17 (multi-stakeholder collaboration across port logistics chains).
- Energy systems: The Port of Livorno incorporates real-time energy consumption monitoring [3], supporting energy planning and aligning port operations with regional environmental goals. This application field links to SDG 7 (clean and efficient energy use), SDG 11 (urban energy resilience), and SDG 12 (responsible energy consumption patterns).

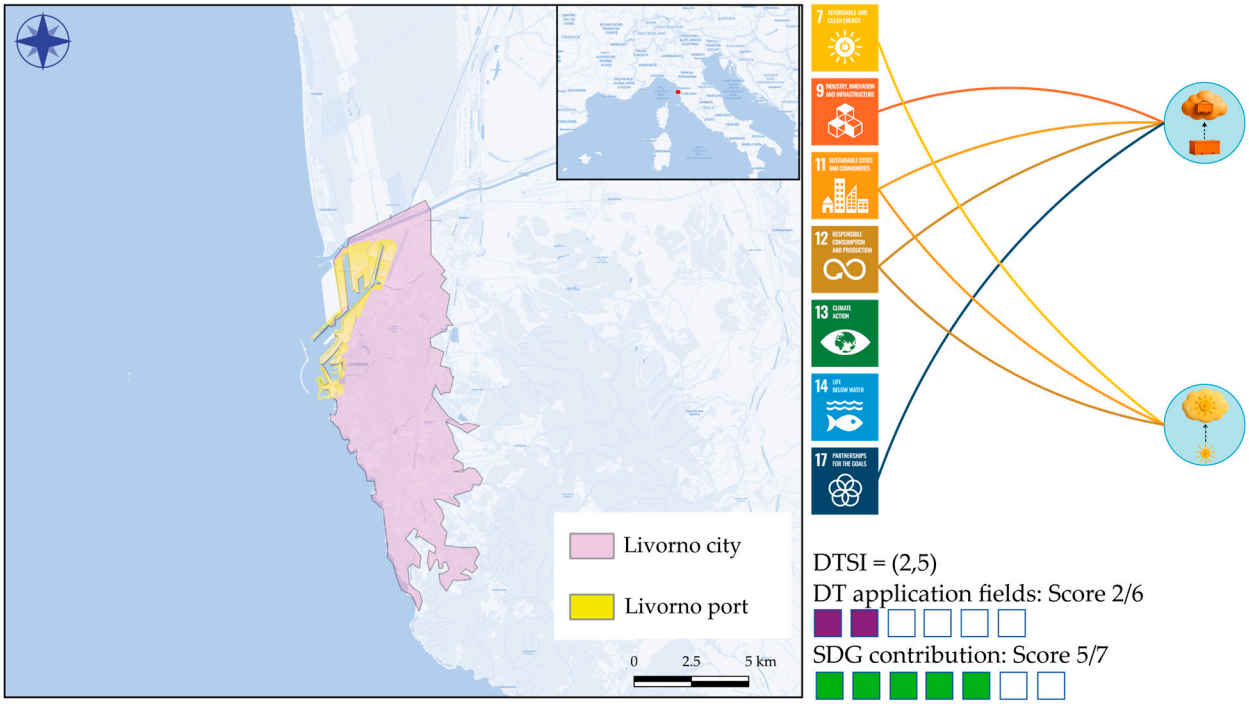


Figure 9. Digitized layout of the Port of Livorno and its urban interface—DT application fields and their connection with SDGs.

Livorno's DT strategy is embedded within its Port Master Plan, which also promotes intermodal transport integration and environmental performance tracking [47]. Through these systems, the port strengthens its capabilities to support regional climate and mobility objectives of the broader urban environment. In summary, the Port of Livorno demonstrates activity in two of the six identified DT application fields and contributes to five of the seven SDGs considered in this study.

The Port of Rotterdam exemplifies a forward-thinking approach to digital transformation, with DT applications supporting both operational excellence and broader urban sustainability. As one of the world's busiest and most technologically advanced ports, Rotterdam's DT initiatives focus on optimizing vessel navigation and container logistics [19]. The key DT application fields are (Figure 10):

- Port traffic and vessel management: Rotterdam has implemented real-time monitoring and predictive routing tools such as hydro-meteo dashboards and digital route planners to support vessel navigation [3]. These systems help determine optimal berthing sequences, reduce waiting times, and ensure smooth port entry [19,38]. Future-oriented tools also support autonomous vessel operations [38] and anticipate water depth changes for safe navigation [68]. This application field contributes to SDG 9 (through intelligent maritime infrastructure and autonomous systems), SDG 12 (delays and fuel consumption reduction), SDG 13 (emission reductions linked to efficient routing), SDG 14 (safe navigation and minimized marine disruption), and SDG 17 (enhancing coordination among maritime stakeholders).
- Logistics, cargo handling, and transportation: Rotterdam's DT framework includes full digital replication of terminal operations and container flows [17]. This can enhance terminal productivity and strengthen community-wide coordination through shared platforms. This application field supports SDG 9 (innovation in multimodal logistics), SDG 11 (efficient freight mobility in the urban region), SDG 12 (reduced energy during cargo handling), and SDG 17 (interoperability across supply chain actors).

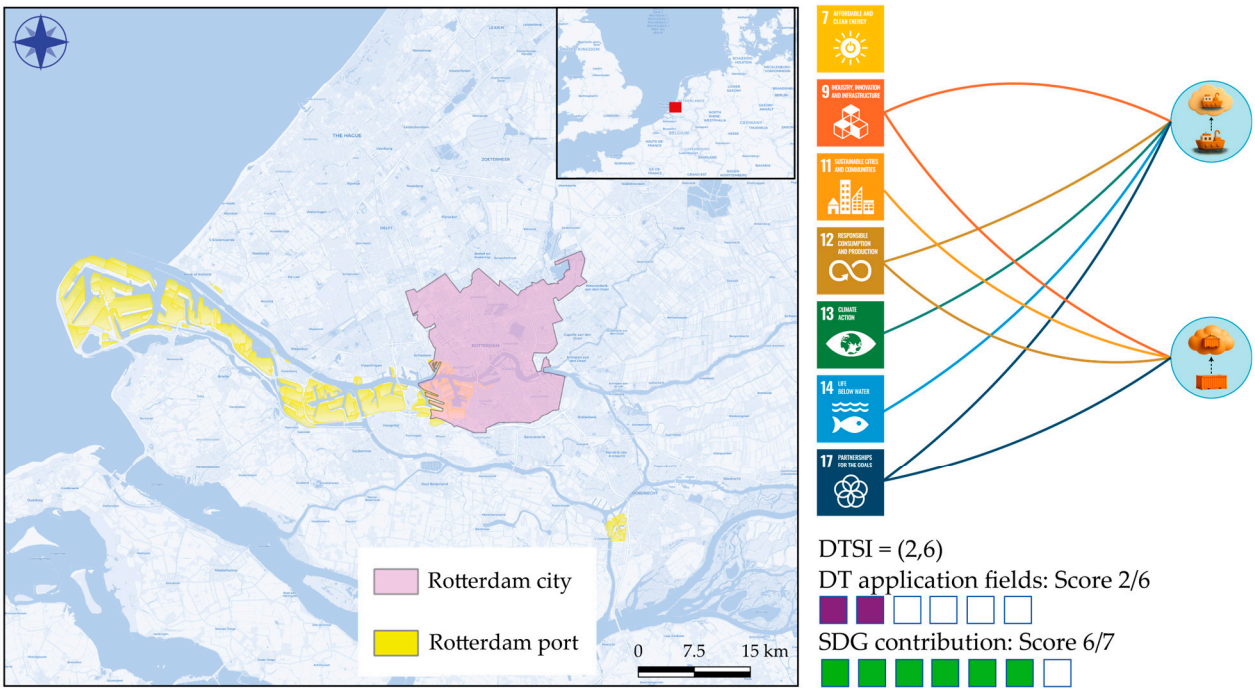


Figure 10. Digitized layout of the Port of Rotterdam and its urban interface—DT application fields and their connection with SDGs.

Rotterdam's DT initiatives can reinforce the city's sustainability ambitions by improving transport connectivity, reducing environmental footprints, and enabling data-driven urban–port cooperation. In summary, the Port of Rotterdam demonstrates activity in two of the six identified DT application fields and contributes to six of the seven SDGs considered in this study.

Finally, the Port of Valencia primarily focuses on enhancing the efficiency and coordination of cargo logistics through DT applications. The key DT application fields are (Figure 11):

- Logistics, cargo handling, and transportation: Valencia implements DT systems to monitor and optimize cargo flow, improving the efficiency of container handling, storage, and multimodal transfer across its facilities [29]. These tools contribute to faster operations, reduced congestion, and better alignment with hinterland distribution. This application field is aligned with SDG 9 (innovative technologies in freight logistics), SDG 11 (improved port–city freight coordination), SDG 12 (streamlining cargo processes), and SDG 17 (strengthening digital collaboration among supply chain actors).
- Environmental monitoring: TwinShip is launched to support emissions reduction in shipping operations and develop an open-source digital twin framework [55]. This application field is linked to SDG 11 (urban environmental quality), SDG 14 (marine protection), and SDG 17 (for potential collaborative innovation ecosystems with start-ups and other partnerships).

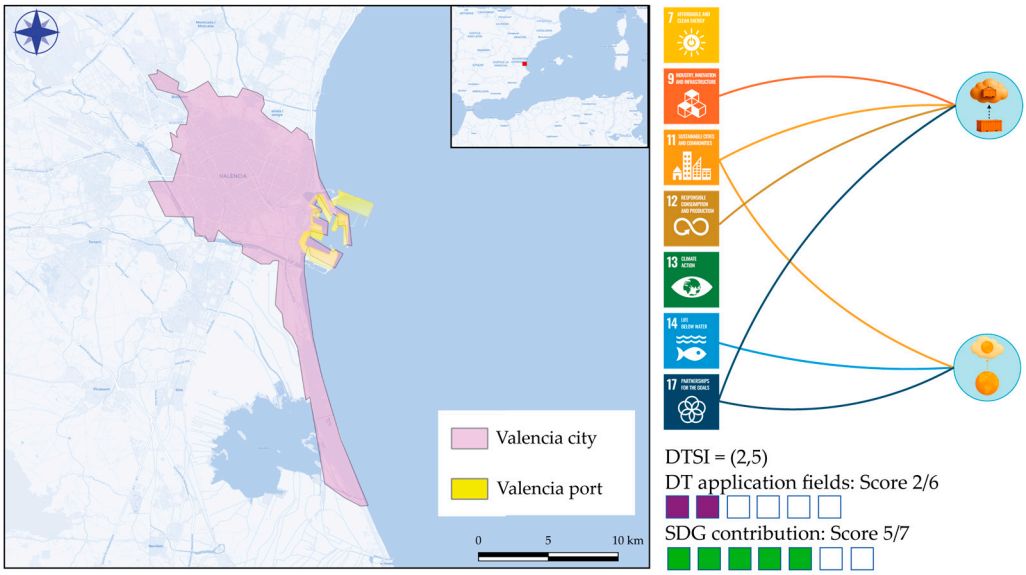


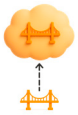
Figure 11. Digitized layout of the Port of Valencia and its urban interface—DT application fields and their connection with SDGs.

Given Valencia's proximity to densely populated urban zones and its economic role in the broader Spanish hinterland, this DT innovation, although in one field, contributes significantly to regional sustainability, efficient mobility, and port–city interoperability. In summary, the Port of Valencia demonstrates activity in two of the six identified DT application fields and contributes to five of the seven SDGs considered in this study.

Table 3 summarizes the DT initiatives implemented across the six European ports, highlighting their alignment with the SDGs. Among these initiatives, cargo management and logistics emerge as the most widely adopted application fields, implemented in all six ports. Port traffic and vessel management follow, with active DT deployment in three of the ports. In contrast, energy systems, despite their critical role in advancing sustainability, have only been integrated in Antwerp-Bruges and Livorno. Infrastructure monitoring, environmental quality tracking, and control and operations centers represent the least developed DT fields, currently implemented only in the Port of Hamburg.

Table 3. DT application fields across six European ports and their corresponding contributions to selected SDGs.

DT Application Field	Involved Ports						SDGs' Explanatory
	Antwerp-Bruges	Barcelona	Hamburg	Livorno	Rotterdam	Valencia	
Port traffic/Vessel management	√		√		√		Innovation in vessel traffic coordination through smart infrastructure
							Reducing unnecessary waiting times and port congestion
							Lowering emissions through efficient vessel flow
							Enhancing marine safety by monitoring ship movements
							Engaging stakeholders such as ship operators through shared data platforms
	√	√	√	√	√	√	Technological innovation in multimodal freight handling

 Cargo handling & management					Improved logistics integration in the urban–port interface
					Reduction in resource waste and more efficient operations
					Strengthening partnerships with cargo owners and logistics actors
 Energy systems	√		√		Support for clean energy use and integration
					Sustainable port–city energy transitions
					Optimized and responsible energy consumption
 Infrastructure condition & maintenance		√			Delivering and maintaining resilient infrastructure
					Improving connectivity with urban networks
					Promoting resource-efficient maintenance
					Increasing resilience to climate impacts through targeted interventions
 Environmental monitoring	√	√		√	Improving urban–port coordination towards addressing pollution
					Protection of marine biodiversity with warning systems
					Strengthening institutional and digital partnerships
 Control operation centers		√			Real-time monitoring in port–city areas
					Strengthening institutional and digital partnerships

3.2. Prospective Pathways for DT Integration at the Port of Piraeus

To assess the potential of DT applications at the Port of Piraeus, the port's spatial layout and its urban interface were digitized in a GIS environment (Figure 12). While the port primarily serves the Piraeus region, its functional influence extends across the wider Athens metropolitan area, confirming a strong urban–port interconnection. Digitization facilitates a clearer understanding of these interdependencies, boundary overlaps, and shared mobility zones between the port and the urban fabric. Despite Piraeus being highly engaged in sustainability efforts and active in innovative European projects, DT deployment remains largely underexplored. Lessons learned from other European ports can be effectively translated to support the digital upgrading of Piraeus, reinforcing both its competitiveness and its alignment with global port trends. These insights are valuable not only for Piraeus but also for other ports aiming to transition into the digital era.



Figure 12. Digitized layout of the Port of Piraeus and its urban interface.

The comparative analysis across Europe highlighted cargo logistics and supply chain optimization as the most prioritized DT application field. This reflects the strategic importance and profitability of freight operations, making them a key area for technological investment. For Piraeus, the integration of DT in cargo logistics could significantly enhance sustainability in the broader urban context by improving freight mobility through better coordination of rail and truck transport, optimizing route planning, and minimizing urban congestion. Given the port's deep connection with the wider transport network, adopting tested logistics solutions, similar to those already implemented in ports like Antwerp or Rotterdam, could offer immediate and scalable benefits. Additionally, establishing partnerships with ports already applying such systems could accelerate the learning curve and integration process.

The second highest-ranking DT field, port traffic and vessel management, is particularly relevant for Piraeus, which is not only a commercial hub but also a critical gateway for domestic ferry services and island connectivity. During the peak tourist season, Piraeus handles large volumes of passenger and cruise traffic. Applying DTs in this field, such as route prediction, berthing optimization, and vessel scheduling, could help reduce congestion, limit delays, and enhance passenger disembarkation efficiency. These benefits would positively influence the adjacent urban environment by easing traffic pressures and improving the overall flow between the port and the city.

Piraeus has already demonstrated a commitment to environmental monitoring, including air and water quality assessment, and ship waste management systems. These efforts create a solid foundation for expanding into digital environmental monitoring, a DT category that could be further developed through sensor integration and predictive modeling. Such initiatives would meet sustainability targets and enhance transparency and reporting in alignment with EU environmental directives.

The remaining DT application areas, i.e., energy systems, infrastructure condition monitoring and maintenance, and control and operations centers, currently appear less mature at the European level but hold long-term value for Piraeus. These areas can be gradually developed through targeted knowledge exchange and collaboration under EU-funded research frameworks such as Horizon Europe, CEF, or Interreg. Leveraging such partnerships could help Piraeus build internal expertise and pilot advanced digital tools aligned with emerging best practices.

In conclusion, a phased DT implementation strategy is proposed for the Port of Piraeus:

- Short-term priorities should focus on cargo logistics and vessel traffic management since these fields are both operationally critical and well-tested in European settings.
- Medium-term actions could strengthen environmental monitoring using digital technologies, building upon the port's existing ESG commitments [62].
- Long-term objectives may include the deployment of DT systems for infrastructure maintenance, energy optimization, and control center integration, supported by European collaborations.

Such a roadmap ensures that the Port of Piraeus progresses toward becoming a future-ready smart port, while simultaneously enhancing the sustainability and resilience of the surrounding urban environment.

For the successful integration of DT technologies, the active involvement of diverse stakeholders is essential. At the Port of Piraeus, key actors include the port authority (PPA S.A.), local and national government entities, municipal authorities, logistics companies, ferry and cruise operators, as well as research institutions and technology providers. Collaboration among these groups is crucial not only for aligning priorities and resources but also for ensuring the interoperability and effectiveness of DT systems across different operational layers. Drawing inspiration from European examples, Piraeus could benefit from structured multi-stakeholder platforms to co-design DT solutions, pilot test new systems, and share data in a secure and transparent way. Such partnerships would accelerate technological uptake and foster shared ownership of sustainability goals, reinforcing the urban–port symbiosis and enabling more resilient and inclusive development.

4. Conclusions

Port–city interfaces represent complex spatial and operational environments, shaped by the different priorities of urban life and maritime trade. Yet, a unifying objective has emerged in recent years: advancing sustainability and delivering climate resilience. In this context, innovative strategies and digital technologies are being explored, not only to support environmental goals but also to enhance competitiveness, operational efficiency, and long-term growth. One such transformative innovation is the use of DT systems across port infrastructures, environmental conditions, processes, and operations.

This research has demonstrated that port DTs are closely aligned with global sustainability agendas, offering tangible benefits to both port performance and the sustainability of the broader urban environments they serve. More broadly, the study offers transferable knowledge for other urban ports by applying a "partial-to-whole" perspective, linking port-level digital innovation to the broader sustainability of urban systems. This complements other studies that follow a "whole-to-part" approach, where smart city strategies inform port upgrades [15]. Bridging these two scales is essential for integrated, future-ready planning.

Through a synthesis of six European case studies, the analysis connected DT application fields with specific SDGs, highlighting how DT systems support climate action, en-

ergy optimization, digital innovation, and responsible consumption. The cross-case synthesis revealed that cargo management and logistics is the most widely implemented DT application field, followed by port traffic and vessel management. These two categories form the foundation for mature DT practices, offering proven pathways for optimization and sustainability. Other application areas, such as energy systems, infrastructure monitoring, control centers, and environmental quality assessment, remain underexplored in most ports, offering space for future innovation and experimentation.

Among the six ports analyzed, Hamburg emerged as the most digitally mature, having implemented DTs in five of the six identified fields. It is followed by Antwerp-Bruges, Rotterdam, Livorno, and finally Barcelona and Valencia, which show a more focused scope. Of particular note is Hamburg's leadership role in co-founding the international chainPORT alliance, an initiative promoting collaborative innovation between global ports, including some of the assessed in this study. Such networks present valuable opportunities for knowledge transfer and co-development of smart and digital port solutions.

The case of the Port of Piraeus shows strong potential for DT transformation. Although not yet fully digitally integrated, Piraeus has established the necessary foundations, especially in sustainability reporting, environmental monitoring, and participation in European research projects. This positions it well to prioritize DT implementations in cargo logistics and vessel management, while gradually building capacity in more advanced fields.

Limitations of this research include the exclusion of pilot projects and conceptual frameworks that have not yet reached full implementation. The focus was deliberately placed on documented and applied DT strategies to offer evidence-based insights. Future research could expand this scope by incorporating pilot initiatives, as well as integrating stakeholder engagement, for example, through surveys, interviews, or roundtable discussions. Such engagement could help prioritize DT pathways, ensuring alignment with local port needs and institutional readiness. Indicatively, in the case of Piraeus, this might reveal that environmental monitoring should be the first DT priority, building on existing practices and sensor infrastructure.

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